NEWTON SCHOOL OF TECHNOLOGY
Data Visualization & Analytics | Capstone 2

Optimizing Hospital Outcomes: Hospital Readmission Rate Analysis

Identifying Risk Factors & Strategic Recommendations for Diabetic Patient Care

Diabetes 130-US Hospitals Dataset · 1999–2008 · UCI Machine Learning Repository

Sector	Healthcare Analytics
Team ID	SEC-A, Group 3 (SEC-A_g-3)
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Project Team

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1. Executive Summary

This report analyses hospital readmission patterns for diabetic patients across 130 U.S. hospitals spanning a nine-year period (1999–2008). Starting from 101,766 raw encounters, the dataset was reduced to 71,515 unique patient records after rigorous cleaning and deduplication. The primary metric of concern — the 30-day readmission rate of 8.8% (approximately 6,300 patients) — is the focal Centre for Medicare & Medicaid Services (CMS) penalty metric, and is the central subject of all intervention strategies developed in this analysis.

71,515	8.8%	39.9%	4.3 days	7.25	18.2%
Total Encounters	30-Day Readmit	Overall Readmit	Avg Stay	Avg Diagnoses	HbA1c Coverage

Key Findings

- 30-day readmission rate stands at 8.8%; overall (any-time) readmission at 39.9%.
- Prior inpatient visits are the single strongest predictor of 30-day readmission ($r = 0.08$).
- Emergency admissions carry significantly higher readmission risk versus non-emergency ($p < 0.05$).
- Lab testing is critically underutilised: HbA1c tested in only 18.2% of encounters; glucose serum in 4.8%.
- High utilisers (>3 prior visits, 3.79% of dataset) face dramatically elevated readmission odds.
- Racial disparities exist: African American patients exhibit notably higher readmission rates.

Key Recommendations

- Mandate HbA1c screening for all diabetic admissions — projected 11% reduction in readmission odds at minimal cost.
- Implement a tiered care-management programme for high utilisers with 48-hour post-discharge follow-up contacts.
- Deploy a real-time readmission risk score at admission, using prior inpatient history, comorbidity count, and age, to triage discharge resources to the patients who most need them.

2. Sector & Business Context

Sector Overview

Hospital readmissions represent one of the most expensive and preventable challenges in modern healthcare. In the United States, Medicare alone spends over USD 26 billion annually on potentially avoidable readmissions. The CMS Hospital Readmissions Reduction Program (HRRP), introduced under the Affordable Care Act, penalises hospitals financially when risk-adjusted 30-day readmission rates exceed national benchmarks for targeted conditions — a category that prominently includes diabetes-related complications.

The 130 hospitals included in this dataset represent a cross-section of U.S. inpatient facilities capturing clinical encounters across a decade of diabetic care. This dataset offers a rare longitudinal view into patient-level encounters, medication regimens, laboratory testing patterns, and discharge dispositions — precisely the variables that drive readmission decisions.

Decision-Maker This Project Serves

The primary stakeholder is the Chief Medical Officer (CMO) or Director of Quality & Patient Safety at a hospital network. Secondary stakeholders include: discharge planning coordinators responsible for post-acute care routing; pharmacy and clinical operations teams managing medication adjustment protocols; and population health managers overseeing community health equity programmes.

Why This Problem Was Chosen

Diabetic patients disproportionately consume hospital resources due to multi-system comorbidities, medication complexity, and the chronic progressive nature of their condition. The CMS financial penalty framework makes readmission rates a direct revenue concern, not merely a clinical quality metric. A meaningful reduction in 30-day readmissions translates simultaneously into improved patient outcomes, reduced CMS penalties, and better resource utilisation.

Business Value of Solving It

A 1-percentage-point reduction in the 30-day readmission rate across a 71,515-patient population translates to approximately 715 fewer readmissions. At an average cost of USD 14,400 per diabetic readmission, this represents potential savings of over USD 10 million per cohort cycle. For a hospital system subject to CMS penalties, even a modest reduction in excess readmission ratios can eliminate five- to six-figure annual penalty charges.

3. Problem Statement & Objectives

Formal Problem Definition

High hospital readmission rates among diabetic inpatients signal systemic gaps in discharge planning, post-acute care coordination, and inpatient clinical protocol adherence. This project investigates: which demographic, clinical, and operational factors most reliably predict 30-day readmission for diabetic patients, and what data-driven interventions can hospital systems implement to reduce this rate.

Project Scope

In Scope

- All 71,515 unique first-encounter diabetic patient records from the UCI Diabetes 130-US Hospitals dataset.
- Demographic analysis (age, race, gender) and its relationship to readmission.
- Clinical factor analysis: length of stay, number of diagnoses, prior utilisation, medication adjustments, and lab test coverage.
- Statistical testing: Chi-squared, Mann-Whitney U, Logistic Regression, and Proportions Z-Test.
- Risk stratification framework for use at point of admission.
- Tableau dashboard for executive and operational decision-making.

Out of Scope

- Predictive machine-learning models (e.g., Random Forest, XGBoost) — noted as future scope.
- Real-time integration with hospital information systems.
- Encounters beyond first admission per patient (to prevent data leakage).
- Analysis of hospitals outside the 130-facility dataset.

Success Criteria

- Identify at least five statistically significant predictors of 30-day readmission ($p < 0.05$).
- Construct a risk stratification framework validated against dataset statistics.
- Produce at least five actionable, measurable recommendations mapped to quantified expected impact.
- Build a Tableau dashboard covering both executive KPI summary and operational drill-down views.

4. Data Description

Dataset Source

Source: UCI Machine Learning Repository — Diabetes 130-US Hospitals for Years 1999–2008.

Citation: Strack, B., DeShazo, J.P., Gennings, C., et al. (2014). Impact of HbA1c Measurement on Hospital Readmission Rates: Analysis of 70,000 Clinical Database Patient Records. BioMed Research International.

Direct Access: https://archive.ics.uci.edu/ml/datasets/Diabetes+130-US+hospitals+for+years+1999-2008

Data Structure

Attribute	Value
Raw Records	101,766 encounters across 130 hospitals
Post-Cleaning Records	71,515 unique patient encounters
Columns (Raw)	50 attributes
Time Period	1999–2008 (9 years)
Target Variable	readmitted (<30 days binary)
Key Variable Groups	Demographics, clinical metrics, lab tests, medications, discharge

Key Column Descriptions

Column	Description
encounter_id	Unique identifier for each inpatient encounter.
patient_nbr	Unique patient identifier; used to extract first encounter per patient.
race	Patient racial identity: Caucasian, African American, Hispanic, Asian, Other.
gender	Patient gender.
age	Age encoded in 10-year bands: [0-10), [10-20), … [90-100).
time_in_hospital	Number of inpatient days (1–14).
num_lab_procedures	Number of laboratory procedures performed.
num_procedures	Number of non-lab procedures.
num_medications	Number of distinct medications administered.
number_outpatient	Prior outpatient visits in preceding 12 months.
number_emergency	Prior emergency visits in preceding 12 months.
number_inpatient	Prior inpatient visits — strongest predictor in the dataset.
number_diagnoses	Number of diagnoses recorded at discharge.
max_glu_serum	Maximum glucose serum test result (None, Norm, >200, >300).
A1Cresult	HbA1c test result (None, Norm, >7, >8).
diag_1 / diag_2 / diag_3	ICD-9 primary, secondary, tertiary diagnosis codes.
readmitted	Target: <30 (30-day readmit), >30 (readmit beyond 30 days), NO.
discharge_disposition_id	Where the patient was discharged: home, SNF, rehab, etc.

admission_type_id	Emergency, Urgent, Elective, Newborn, etc.
insulin	Insulin dosage change: No, Steady, Up, Down.

Data Limitations & Known Biases

- Historical dataset (1999–2008): clinical protocols have evolved substantially; findings may not perfectly generalise to contemporary practice.
- Weight column: 96.9% missing — cannot be used for analysis or imputation.
- Payer code: 39.6% missing — excluded from analysis; insurance-type effects cannot be assessed.
- Near-zero variance in most medication columns (>99% 'No' entries): most drug features carry no signal for readmission prediction.
- ICD-9 coding is used rather than the more current ICD-10 system.
- Racial categories are self-reported and may not capture full demographic nuance.
- Dataset represents 130 hospitals in the U.S. only — results may not generalise to other healthcare systems.

5. Data Cleaning & ETL Pipeline

All primary cleaning and transformation steps were executed in Python (Pandas, NumPy) as per Capstone 2 requirements. Reference notebook: notebooks/02_cleaning.ipynb committed to GitHub.

Missing Value Treatment

Column	Missing %	Treatment
weight	96.9%	Dropped — imputation indefensible at this rate.
payer_code	39.6%	Dropped — not clinically relevant to readmission prediction.
race	~2.2%	Retained as 'Unknown' category; tested separately in analysis.
medical_specialty	~49%	Retained; treated as 'Unknown/Missing' for specialty-level analysis.

Deduplication

The raw dataset contains multiple encounters per patient. To prevent data leakage and ensure each patient's readmission outcome is treated independently, only the first recorded encounter per patient_nbr was retained. This reduced the dataset from 101,766 to 71,515 records.

Outlier Detection & Treatment

- time_in_hospital and num_medications were inspected for outliers using IQR-based detection. No records were dropped; extreme values were retained as clinically valid (long-stay patients are real and clinically meaningful).
- Discharge disposition codes corresponding to death, hospice, or expired were excluded from readmission analysis, as these patients cannot be readmitted.

Data Type Conversions & Standardisation

- Age bands (e.g., '[60-70)') were converted to numeric midpoints (e.g., 65) for regression and correlation analysis — stored as age_midpoint.
- readmitted column was binarised: '<30' = 1 (30-day readmission), all others = 0 — stored as readmitted_binary.
- ICD-9 diagnosis codes (diag_1, diag_2, diag_3) were grouped into 18 clinical categories (Circulatory, Respiratory, Digestive, Diabetes, Musculoskeletal, etc.) — stored as diag_1_category.

Feature Engineering

Engineered Feature	Description
readmitted_binary	Binary 30-day readmission target (1 = readmitted <30 days).
total_prior_visits	Sum of number_outpatient + number_emergency + number_inpatient.
high_utiliser	Flag: 1 if total_prior_visits > 3, else 0 (3.79% of dataset).
diag_1_category	ICD-9 primary diagnosis mapped to one of 18 clinical categories.
a1c_test_performed	Binary: 1 if A1Cresult is not 'None', else 0.
age_midpoint	Numeric midpoint of 10-year age band for quantitative analysis.

Assumptions Made During Cleaning

- First encounter per patient is considered the most clinically informative and uncontaminated observation for predicting readmission.
- Medication columns with >99% 'No' entries are treated as having no signal and are excluded from statistical analysis.
- Missing racial category is treated as a distinct group, not imputed, to avoid attributing outcomes to a race.

6. KPI & Metric Framework

Reference: notebooks/03_kpi.ipynb committed to GitHub. All KPIs are computed in Python and verified against dashboard outputs in Tableau.

KPI	Formula / Computation	Business Rationale	Objective Mapping
30-Day Readmission Rate	Count(readmitted_binary=1) / Total Encounters	Primary CMS penalty metric; drives hospital revenue risk.	Primary objective
Overall Readmission Rate	Count(readmitted ≠ 'NO') / Total Encounters	Captures all repeat hospitalisations; broader quality signal.	Secondary objective
Avg Length of Stay	Mean(time_in_hospital)	Proxy for clinical complexity; correlated with readmission.	Clinical analysis
Avg Diagnoses / Patient	Mean(number_diagnoses)	Comorbidity burden; predicts care complexity at discharge.	Risk segmentation
Avg Medications / Patient	Mean(num_medications)	Medication complexity; risk factor for non-adherence.	Clinical analysis
HbA1c Coverage Rate	Count(a1c_test_performed=1) / Total Encounters	Measures adherence to diabetes management best practice.	Recommendation impact
Glucose Test Coverage	Count(max_glu_serum≠'None') / Total Encounters	Lab testing adherence; associated with readmission reduction.	Recommendation impact
High Utiliser Rate	Count(high_utiliser=1) / Total Encounters	Identifies highest-cost, highest-risk patient cohort.	Care management

7. Exploratory Data Analysis

Reference: notebooks/03_eda.ipynb committed to GitHub. All charts are reproduced in the Tableau dashboard.

7.1 Demographic Analysis

Age Distribution & Readmission

Patients aged 50–80 represent both the highest readmission rates AND the largest patient volumes in this dataset — making them the optimal target for intervention. The 70–90 age band exhibits peak readmission rates across all racial groups. Patients below 30 and above 90 represent small cohorts with variable rates that do not justify universal protocol changes.

Business implication: Discharge planning protocols should be disproportionately allocated to the 50–80 age band. A targeted 'older adult diabetes discharge checklist' could capture the majority of preventable readmissions.

Racial Disparities in Readmission

African American patients exhibit notably higher 30-day readmission rates compared to Caucasian patients, even when controlling for clinical factors. This points to disparities in post-discharge support, social determinants of health (housing stability, food security, transport access), and potentially implicit bias in discharge care intensity.

Business implication: A community health intervention programme specifically targeting African American diabetic patients is warranted, including social work integration at discharge, community health worker follow-up, and culturally tailored diabetes education.

Gender Analysis

Differences in readmission rates between male and female patients are minimal and not statistically significant at a clinically actionable level. Gender-specific protocols are not justified by current evidence. Analytical resources should not be devoted to this dimension.

7.2 Clinical Factor Analysis

Length of Stay

Readmitted patients have a statistically significantly longer average stay (4.58 days) compared to non-readmitted patients (4.24 days), with a modest effect size ($r = 0.03$, $p < 0.05$). This suggests readmitted patients present with greater clinical complexity requiring extended inpatient management.

Business implication: Discharge planning should be triggered proactively for patients approaching 7+ days of inpatient stay, not reactively at discharge day.

Comorbidity Burden

Readmitted patients carry a meaningfully higher average comorbidity count (7.68 diagnoses) versus non-readmitted patients (7.19 diagnoses), with $r = 0.05$, $p < 0.05$. Patients with 7 or more diagnoses should be automatically flagged in the electronic medical record (EMR) system for enhanced post-discharge coordination.

Prior Healthcare Utilisation

Number of prior inpatient visits is the single strongest utilisation predictor in the dataset, with a mean of 0.31 in readmitted versus 0.16 in non-readmitted patients ($r = 0.08$). High utilisers — defined as patients with more than 3 prior visits and constituting 3.79% of the dataset — face dramatically elevated readmission odds. The Proportions Z-Test confirms this difference is statistically significant ($p < 0.05$).

Business implication: A dedicated care management programme for high utilisers, including assigned care coordinators, transitional care nurses, and 48-hour post-discharge telephone contact, could materially reduce both readmissions and CMS penalties.

7.3 Medication & Treatment Analysis

Insulin Dosage Changes

Insulin dosage reductions ('Down' category) correlate with higher readmission rates, indicating that de-escalation of insulin without structured follow-up monitoring carries elevated risk. Medication adjustments alone are insufficient as an intervention — they must be paired with close post-discharge monitoring.

Lab Test Coverage & Readmission Association

HbA1c testing was performed in only 18.16% of encounters. Among those tested, readmission rates were lower (OR = 0.89, $p < 0.05$), indicating an 11% reduction in readmission odds. Glucose serum testing was performed in only 4.83% of encounters, with a similar protective association observed. These coverage rates represent a significant clinical quality gap.

Business implication: Mandating HbA1c testing for all diabetic admissions is one of the highest-return, lowest-cost interventions available. The incremental cost of an HbA1c test (approximately USD 15–25) is vastly outweighed by the projected readmission cost savings of USD 14,400 per avoided readmission.

7.4 Diagnosis Category & Admission Pattern Analysis

Primary Diagnosis Distribution

Circulatory system disorders are the most frequent primary diagnosis category AND carry above-average readmission rates. Only 8.12% of encounters list diabetes as the primary diagnosis, yet all patients are diabetic — indicating that diabetes frequently presents as a secondary or complicating condition driving the primary admission.

Business implication: Joint cardiac-diabetic discharge protocols are essential for the circulatory cohort. These patients need coordinated cardiology and endocrinology discharge planning.

Admission Type

Emergency admissions drive the highest readmission rates and this difference is statistically significant versus non-emergency admissions ($p < 0.05$). Emergency admissions reflect patients who arrive in a more acute clinical state, with less opportunity for planned discharge coordination.

Discharge Disposition

Patients discharged directly to home have higher readmission rates than those discharged to Skilled Nursing Facilities (SNFs) or rehabilitation centres. This finding identifies home discharge without structured follow-up as an independent, modifiable risk factor.

8. Statistical Analysis

Reference: notebooks/04_statistical_analysis.ipynb committed to GitHub. Four statistical methods were applied to ensure robustness of findings.

8.1 Chi-Squared Tests of Independence

Chi-squared tests were conducted for all categorical variables against the readmitted_binary target to test statistical independence. Most variables showed statistically significant associations ($p < 0.001$). However, effect sizes (Cramér's V) were uniformly below 0.1, which is typical in large healthcare datasets at 70,000+ records where statistical significance is easily achieved but practical effect sizes remain modest.

Variable	p-value	Cramér's V	Relative Effect Magnitude
discharge_disposition_id	< 0.001	Largest in dataset	Highest practical significance
admission_type_id	< 0.001	High	High practical significance
diag_1_category	< 0.001	Moderate	Moderate practical significance
age_group	< 0.001	Moderate	Moderate practical significance
race	< 0.001	Small	Actionable disparity signal
insulin	< 0.001	Small	Clinically actionable
medical_specialty	< 0.001	Small	Context-dependent

8.2 Mann-Whitney U Tests

Mann-Whitney U tests (non-parametric) were used for numeric variables to compare distributions between readmitted and non-readmitted groups, avoiding the normality assumption.

Variable	Mean (Readmit)	Mean (No Readmit)	Effect r	Direction
number_inpatient	0.31	0.16	0.08	Higher in readmitted
number_diagnoses	7.68	7.19	0.05	Higher in readmitted
total_prior_visits	0.72	0.53	0.04	Higher in readmitted
time_in_hospital	4.58	4.24	0.03	Higher in readmitted
num_medications	16.22	15.63	0.02	Higher in readmitted

Key finding: Number of prior inpatient visits has the largest effect size — the single strongest numeric predictor of 30-day readmission across all continuous variables tested.

8.3 Logistic Regression — Key Odds Ratios

A logistic regression model was fitted on the cleaned dataset to quantify the independent contribution of key predictors to 30-day readmission probability.

Feature	Coefficient	Odds Ratio	Interpretation
number_inpatient	+0.31	1.36	Each additional prior visit: +36% odds of readmission.
high_utiliser	+0.25	1.28	High utilisers: 28% elevated readmission odds.
number_diagnoses	+0.08	1.08	Each additional diagnosis: +8% readmission odds.
time_in_hospital	+0.04	1.04	Longer stays marginally increase risk.
a1c_test_performed	−0.12	0.89	HbA1c testing: 11% lower readmission odds (protective).

Prior inpatient visits and high-utiliser status are the dominant modifiable risk factors identified by the regression model.

8.4 Proportions Z-Test

A Proportions Z-Test confirmed that the difference in 30-day readmission rates between high utilisers (>3 prior visits) and the rest of the patient population is statistically significant ($p < 0.05$). This validates the high-utiliser segmentation as a clinically and operationally meaningful stratification criterion.

9. Tableau Dashboard Design

Reference: tableau/screenshots/ and tableau/dashboard_links.md committed to GitHub. Tableau Public URL: [Insert Tableau Public Dashboard URL].

Dashboard Objective

The dashboard supports two levels of decision-making: (1) an Executive Summary view for CMOs and quality directors who need to understand overall readmission performance and priority risk factors at a glance; and (2) an Operational Drill-Down view for discharge planners and care coordinators who need patient-segment-level detail to allocate resources.

View Structure

View 1: Executive Summary Dashboard

- KPI banner: Total encounters, 30-day readmission rate, overall readmission rate, average length of stay, HbA1c coverage rate.
- Donut chart: 30-day vs. non-readmitted split.
- Bar chart: Readmission rates by age group and race.
- Trend line: Readmission rates over time (by year of encounter).
- Filters: Hospital, admission type, primary diagnosis category.

View 2: Operational Drill-Down Dashboard

- Heatmap: Readmission risk by discharge disposition vs. admission type.
- Scatter plot: Number of diagnoses vs. prior inpatient visits (coloured by readmission outcome).
- Bar chart: Readmission rates by primary diagnosis category (ICD-9 group).
- Lab testing gap chart: HbA1c and glucose test coverage rates vs. benchmark.
- Risk tier distribution: Count of patients in each of the four risk tiers (Critical, High, Medium, Low).

Interactive Elements

- Age group filter (multi-select).
- Race filter (multi-select).
- Admission type filter (Emergency / Non-Emergency toggle).
- Discharge disposition filter.
- Primary diagnosis category filter.
- Readmission outcome filter (30-day / >30-day / Not readmitted).

10. Insights Summary

The following 10 insights are written in decision language — each tells the reader what to conclude and, where possible, what action follows.

#	Insight
1	Patients aged 50–80 generate the highest volume AND the highest readmission rates — concentrating discharge planning resources on this cohort will capture the greatest absolute number of preventable readmissions.
2	Each additional prior inpatient visit increases readmission odds by 36% — prior utilisation history is the single most important data point to check at the time of admission.
3	High utilisers (>3 prior visits, 3.79% of patients) represent a small but disproportionately high-risk cohort that justifies dedicated care management investment.
4	Emergency admissions carry significantly higher readmission risk — these patients require enhanced discharge protocols by default, not as an exception.
5	Home discharge without structured follow-up is itself an independent readmission risk factor — the destination of discharge matters as much as clinical readiness.
6	HbA1c testing is performed in only 1 in 5 diabetic patients, despite being associated with an 11% reduction in readmission odds — this is the highest-return, lowest-cost clinical intervention available.
7	Circulatory system disorders are the dominant primary diagnosis driving readmissions — hospitals without a joint cardiac-diabetic discharge protocol are missing a critical intervention point.
8	Patients with 7+ diagnoses are significantly more likely to be readmitted — comorbidity count should trigger automatic enhanced discharge planning flags in the EMR.
9	African American patients show meaningfully higher readmission rates, pointing to systemic inequities in post-discharge support, not just clinical factors — community health interventions are essential.
10	Insulin dose reductions without concurrent monitoring are associated with higher readmission volatility — de-escalation decisions must always be paired with structured post-discharge monitoring protocols.

11. Recommendations

	Recommendation	Expected Business Impact	Feasibility Notes
1	Mandate HbA1c screening for all diabetic inpatient admissions.	11% reduction in readmission odds. At 6,300 baseline readmissions, ~693 readmissions potentially avoided per cohort cycle.	Low cost per test (~USD 15-25). Requires EMR-level order set modification and a clinical policy change. 30–60 day rollout feasible.
2	Implement dedicated care management programme for high utilisers (>3 prior visits) with assigned care coordinators and 48-hr post-discharge telephone contact.	28% lower readmission odds for high utilisers. Targets 3.79% of patients (~2,700 per cohort) with concentrated intervention.	Requires 2–4 FTE care coordinators per 5,000-patient population. Transitional care model is well-documented; pilots can launch within 60–90 days.
3	Deploy a real-time readmission risk score at admission using number_inpatient, number_diagnoses, age, and admission_type as primary inputs.	Enables stratified allocation of discharge resources. Prevents over- and under-resourcing of discharge planning teams.	Requires EMR integration (Epic, Cerner) — typically 3–6 month IT project. Logic can be piloted on a single unit as a paper-based checklist within weeks.
4	Establish joint cardiac-diabetic discharge protocols for patients with circulatory primary diagnoses.	Addresses the highest-volume, above-average readmission risk segment in the dataset.	Requires cross-departmental protocol development between cardiology and endocrinology. 90-day implementation timeline for initial draft protocol.
5	Integrate community health worker support and social work referrals at discharge for African American patients and all home-discharged patients.	Addresses racial disparities and home-discharge risk simultaneously. Long-term reduction in social determinant-driven readmissions.	Requires community health partnership development. Social work integration at discharge can begin immediately with existing staff.

12. Impact Estimation

Impact estimates are based on logistic regression odds ratios and literature benchmarks for intervention effectiveness. Approximate numbers are used — the underlying logic is sound and conservatively stated.

Baseline

71,515 patient encounters. 8.8% 30-day readmission rate = approximately 6,293 readmissions per cohort. Average cost per diabetic readmission: USD 14,400 (AHRQ benchmark). Baseline readmission cost: approximately USD 90.6 million per cohort cycle.

Scenario 1: Mandate HbA1c Screening

- If HbA1c testing coverage increases from 18.2% to 100%, the protective effect (OR 0.89) applies to an additional 81.8% of encounters (~58,500 patients).
- Conservative estimate: 5–7% overall reduction in 30-day readmissions = 315–440 fewer readmissions.
- Cost saving: USD 4.5M – USD 6.3M per cohort cycle.
- Implementation cost: ~USD 15 per test × 58,500 additional tests = USD 877,500.
- Net benefit: USD 3.6M – USD 5.4M per cohort.

Scenario 2: Care Management for High Utilisers

- High utilisers: ~2,710 patients (3.79% of 71,515). Their readmission rate is disproportionately elevated.
- Transitional care programmes in literature reduce readmissions by 15–25% in high-utiliser cohorts.
- At 20% reduction among high utilisers: approximately 80–120 fewer readmissions.
- Cost saving: USD 1.2M – USD 1.7M per cohort cycle.
- Programme cost: 3 FTE care coordinators at ~USD 80,000/year = USD 240,000.
- Net benefit: USD 960K – USD 1.46M per cohort.

Scenario 3: Risk Score Deployment

- Risk scores do not directly reduce readmissions — they enable resource efficiency.
- Eliminating unnecessary enhanced discharge packages for low-risk patients (estimated 20–30% of enhanced packages currently misallocated) frees resources equivalent to USD 500K – USD 1M in discharge team labour per cohort.

Combined Estimated Impact

Implementing all three priority recommendations has a combined projected value of USD 5M – USD 8M in readmission cost avoidance plus efficiency gains per cohort cycle, against an implementation investment estimated at USD 1.2M – USD 1.5M. Return on investment: approximately 4:1 to 5:1.

13. Limitations

Data Quality Constraints

- Weight column (96.9% missing) and payer_code (39.6% missing) could not be used. BMI and insurance type are clinically relevant readmission predictors — their absence limits the completeness of the risk model.
- Medical specialty (49% missing) introduces uncertainty in specialty-level analysis.
- The dataset is historical (1999–2008). Clinical protocols, medication regimens, discharge practices, and CMS penalty structures have all evolved since this period. Findings indicate structural patterns but may not perfectly generalise to a 2024+ clinical context.

Analytical Assumptions

- First-encounter-only design prevents data leakage but means that patients with multiple admissions across the study period are underrepresented relative to their actual utilisation burden.
- Effect sizes (Cramér's V, r) are uniformly small across the dataset, consistent with the complexity of healthcare outcomes. No single variable fully 'explains' readmission — the risk is multi-factorial.
- The logistic regression model does not include interaction terms (e.g., age × race × comorbidities), which may understate the combined impact of overlapping risk factors.

What Cannot Be Concluded

- Causal relationships cannot be established — this is a retrospective observational study. Association between HbA1c testing and lower readmission does not prove that testing itself causes the reduction.
- These findings cannot be directly generalised to non-diabetic patient populations, paediatric hospitals, or healthcare systems outside the United States.
- Social determinant data (housing stability, food security, health literacy) is absent — the racial disparity findings may be substantially driven by social determinants that are unmeasured in this dataset.

14. Future Scope

Advanced Predictive Modelling

The current analysis used logistic regression for odds ratio interpretation. A machine-learning model (Random Forest, Gradient Boosted Trees, or XGBoost) trained on the full feature set — including interaction terms and engineered features — would likely achieve materially higher predictive accuracy (AUROC > 0.70 expected vs. ~0.62 for the current logistic regression). This would enable a patient-level readmission probability score deployable at the point of admission.

Additional Data Sources

- Social determinant data (housing, food security, transport): would substantially improve disparity analysis and community intervention targeting.
- Insurance/payer type: would enable analysis of financial access barriers to post-discharge follow-up.
- Post-discharge follow-up records: would directly test the impact of structured follow-up on readmission outcomes.
- Contemporary clinical data (2015 onwards): would increase relevance to current practice and CMS penalty structures.
- Medication adherence data: would enable analysis of post-discharge non-adherence as a readmission driver.

Real-Time Dashboard & EMR Integration

The current Tableau dashboard operates on static batch data. The logical next step is integration with a hospital EMR system (Epic, Cerner) via API, enabling real-time risk score computation at the point of admission and dynamic updating of discharge planning queues. A live dashboard would also enable measurement of intervention effectiveness over time, creating a feedback loop for continuous protocol improvement.

Natural Language Processing

Clinical notes (unstructured text) represent a rich but underutilised data source. NLP-based feature extraction from discharge summaries and nursing notes could identify readmission risk signals not captured by structured fields.

15. Conclusion

Analysis of 71,515 diabetic patient encounters across 130 U.S. hospitals confirms significant, statistically validated, and operationally actionable patterns in 30-day hospital readmissions. The 8.8% 30-day readmission rate — representing approximately 6,300 patients per cohort cycle and over USD 90 million in associated costs — can be meaningfully reduced by targeting three primary drivers: prior healthcare utilisation, comorbidity burden, and admission type.

HbA1c testing, currently performed for fewer than 1 in 5 diabetic patients, is associated with an 11% reduction in readmission odds and represents the highest-return, lowest-cost clinical intervention available. Alongside mandatory HbA1c screening, a care coordination programme for high utilisers and an enhanced home-discharge follow-up protocol represent the highest-priority actions for any hospital system seeking to reduce CMS penalties, improve patient outcomes, and allocate clinical resources efficiently.

The risk stratification framework developed in this analysis — stratifying patients into Critical, High, Medium, and Low tiers at the point of admission based on prior inpatient history, comorbidity count, age, and admission type — provides hospital teams with a practical, evidence-based tool that can be implemented immediately as a clinical checklist and upgraded progressively into a fully automated EMR-integrated risk score.

Appendix

A. Data Dictionary (Full Column Definitions)

Column	Type	Full Definition
encounter_id	Integer	Unique numeric identifier for each hospital encounter.
patient_nbr	Integer	Unique numeric identifier for each patient across all encounters.
race	Categorical	Patient's self-reported race: Caucasian, AfricanAmerican, Hispanic, Asian, Other, or missing.
gender	Categorical	Patient's gender: Male, Female, or Unknown/Invalid.
age	Categorical	Age in 10-year intervals: [0-10), [10-20), ..., [90-100).
weight	Numeric	Patient's weight in pounds. Dropped (96.9% missing).
admission_type_id	Integer	Type of admission: 1=Emergency, 2=Urgent, 3=Elective, 4=Newborn, 5=Not Available, 6=NULL, 7=Trauma, 8=Not Mapped.
discharge_disposition_id	Integer	Discharge routing code: 1=Home, 3=SNF, 6=Home Health, 11=Expired, etc.
admission_source_id	Integer	Source of the admission: physician referral, emergency room, transfer, etc.
time_in_hospital	Integer	Number of days between admission and discharge (1–14).
payer_code	Categorical	Insurance type code. Dropped (39.6% missing; clinically irrelevant).
medical_specialty	Categorical	Medical specialty of the admitting physician. ~49% missing.
num_lab_procedures	Integer	Number of laboratory tests performed during encounter.
num_procedures	Integer	Number of non-laboratory medical procedures performed.
num_medications	Integer	Number of distinct generic medications administered.
number_outpatient	Integer	Number of outpatient visits in the preceding 12 months.
number_emergency	Integer	Number of emergency visits in the preceding 12 months.
number_inpatient	Integer	Number of inpatient visits in the preceding 12 months. Strongest readmission predictor.
diag_1	Categorical	Primary ICD-9 diagnosis code.
diag_2	Categorical	Secondary ICD-9 diagnosis code.
diag_3	Categorical	Tertiary ICD-9 diagnosis code.
number_diagnoses	Integer	Number of diagnoses entered into the system.
max_glu_serum	Categorical	Result of glucose serum test: None, Norm, >200, >300.
A1Cresult	Categorical	HbA1c test result: None (not tested), Norm, >7, >8.
insulin	Categorical	Insulin dosage change during encounter: No, Steady, Up, Down.
readmitted	Categorical	Readmission outcome: <30 (within 30 days), >30 (after 30 days), NO (not readmitted).

B. Risk Stratification Framework

Risk Tier	Criteria	Recommended Action	Escalation Trigger

Critical	Prior inpatient ≥ 2 OR emergency admission	Intensive case management + 48-hr post-discharge contact	Same-day social work referral
High	Age 70–80 OR ≥ 7 diagnoses OR high utiliser flag	Enhanced discharge planning + care coordinator assignment	Review by discharge team lead
Medium	Age 50–70 OR 4–6 diagnoses	Standard follow-up protocol + HbA1c mandate	Standard nursing follow-up
Low	None of above criteria met	Routine discharge + general diabetes education	None required

C. Statistical Test Reference

Chi-Squared Test: Tests whether two categorical variables are statistically independent. Used for all categorical predictor vs. readmitted_binary pairs. Significant result ($p < 0.05$) means the variables are associated.

Cramér's V: A standardised effect size measure for chi-squared tests, ranging from 0 (no association) to 1 (perfect association). Values below 0.1 indicate small but often practically meaningful associations in large datasets.

Mann-Whitney U Test: Non-parametric test comparing distributions of a continuous variable between two groups. Does not assume normality. Used for all numeric predictor vs. readmitted_binary pairs.

Effect size r: Standardised effect size for Mann-Whitney U, ranging from 0 to 1. Conventionally: 0.1 = small, 0.3 = medium, 0.5 = large.

Logistic Regression: Predicts the probability of a binary outcome (readmitted_binary) as a function of multiple predictors. Odds ratios > 1 indicate increased readmission risk; < 1 indicate protective effects.

Proportions Z-Test: Tests whether the readmission rate in a specific subgroup (e.g., high utilisers) is significantly different from the rest of the population.

Contribution Matrix

This section documents each team member's contribution across all project phases. Claims match evidence in GitHub Insights, PR history, and committed files.

Team Member	Data Sourcing	ETL & Cleaning	EDA & Analysis	Statistical Analysis	Tableau Dashboard	Report Writing	PPT & Viva
Suryansh Singh	Owner	-	-	-	Owner	-	-
Pankaj Baid	Owner	Owner	Owner	-	-	-	-
Pranjal Modi	-	Owner	-	-	-	-	-
Aditya Kishore Singh	-	-	-	Owner	-	Owner	-
Vaibhav	Owner	-	-	-	-	Owner	-
Avishkar Meher	Owner	-	-	-	-	-	-
Saksham Sharma	-	-	-	-	-	-	-

Declaration: We confirm that the above contribution details are accurate and verifiable through GitHub Insights, PR history, and submitted artefacts.

Team Lead Signature: _____

Date: _____